

TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

TECHNICAL REPORT

Field	Details
Institute / University	Pimpri Chichwad College Of Engineering, Pune
Programme / Batch	TechPulse FY-26
Student Name	Sarthak Shiavaji Bagul
PRN	124B2B021
Case Study ID (CS1–CS5)	CS1
Project Title	AUTOSAR HLD Document Analysis Assistant

1. Problem Statement and Intended Users

Classic AUTOSAR High-Level Design (HLD) documents define component boundaries, ports, Sender-Receiver and Client-Server interfaces, signals, and timing constraints. They are long, table-heavy, and reviewed manually, which is slow and error-prone. During Engineering Change Notices (ECNs), breaking changes such as renamed signals, removed ports, data type mismatches, and orphaned interfaces often go unnoticed. They then surface as costly integration bugs in HIL testing or vehicle integration.

Intended Users

1. Systems Architects and Software Leads: design reviews and interface compatibility checks across revisions.
2. Embedded Software Engineers: querying interfaces, execution periods, and signal mappings.
3. Test and Validation Engineers: deriving HIL test cases from functional flows.
4. Quality and Governance Auditors: traceability audits using cited, logged answers.

2. Input Data / Knowledge Base

Item	Description
Sources	Synthetic, script-generated AUTOSAR HLD documents (fixed-seed ReportLab 4.1.0, MIT licence): 1. HLD_BodyControl_v1.pdf, the baseline Body Control Module ECU specification (modelled on Classic AUTOSAR 4.4.0). 2. HLD_BodyControl_v2.pdf, the revision under ECN-2026-08 with 6 seeded discrepancies across 5 categories. 3. seeded_defects_ground_truth.json, listing each defect's ID, category, page and section.
Size and format	Two PDFs of 14 pages each (v1: 58 KB, v2: 60 KB) with headings, tables and flow descriptions. Content: 10 software components, 11 port interfaces (S/R and C/S), 13 signals over CAN-FD Channel 1 and LIN Channel 0, 12 connector pinouts, 2 functional flows, 7 UDS DTC matrices.
Preprocessing / EDA done	1. PyMuPDF extracts text blocks, font sizes and positions to detect sections by page. 2. pdfplumber extracts tables and converts them to Markdown. 3. Regex validation of document IDs, ECN labels and version headers.

Item	Description
	4. Manual inspection checked that no tables were broken and no text fragments were orphaned.
Chunking strategy	Section-aware chunking: 600 characters with 100 overlap, split at H1/H2 headings. Tables stay whole, and when split, the header row is repeated. Each chunk carries doc_id, version, page, section, chunk_type and project_id. Result: 72 chunks for v1 and 75 for v2, stored in ChromaDB.

3. Solution Architecture

Component	Choice (name, version, hosting)
Pre-trained model	Qwen2.5:3b (Q4_K_M quantized), run locally through Ollama on localhost:11434, with no external API calls. Qwen2.5:7b is an option for machines with 16 GB or more RAM.
Embedding model	BAAI/bge-small-en-v1.5 (384-dim), run locally on CPU via sentence-transformers 2.5.1.
Vector store (FAISS / ChromaDB)	ChromaDB 0.4.24, persistent on disk (Code/chroma_db_store), HNSW index with cosine distance. A NumPy cosine fallback is used for small collections.
RAG pipeline / AI-agent design	Top-k = 4 retrieval with optional version filter. Document text is wrapped in <CONTEXT_CHUNKS> tags, separate from instructions (prompt-injection defence). A two-stage refusal gate (cosine similarity ≥ 0.68 , keyword overlap ≥ 0.35) answers "Not found in the provided document" when evidence is weak. A post-check verifies that cited pages were actually retrieved and match the answer. If Ollama is offline, an extractive fallback keeps results reproducible.
Service layer and UI	Streamlit 1.32.2 on localhost:8501 with 6 tabs: Q&A, Ingestion, Entity Explorer, Revision Compare, Export, Audit Log. Banner: "AI-assisted output. Requires engineer review." There is no separate API layer (FastAPI is listed as future work).
Tools / deterministic rules	compare.py has 5 rules: renamed signal, removed port, data type change, undefined component reference, orphaned interface. Detection is fully rule-based, and the LLM only explains the findings.

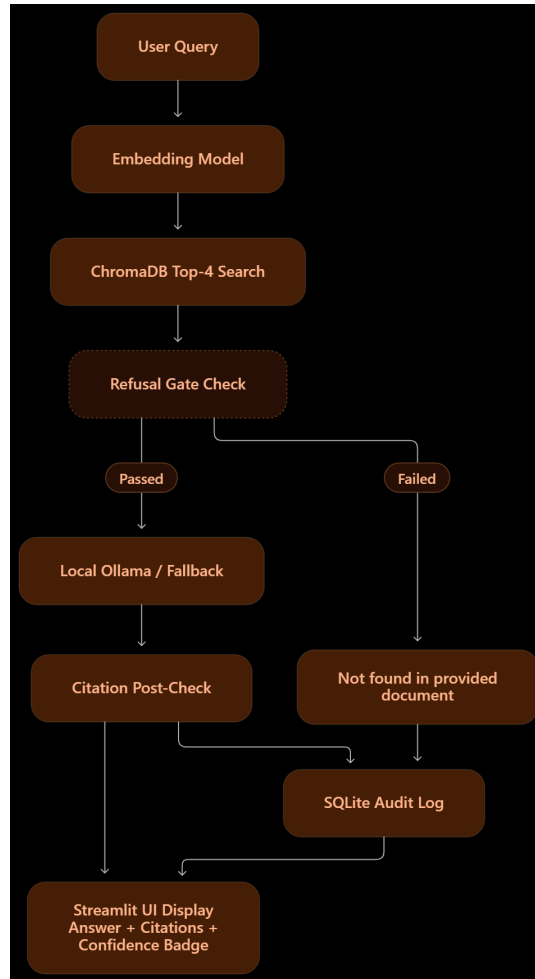


4. Implementation and Configuration

Workflow

1. **Ingestion and parsing.** PyMuPDF extracts page-indexed text, headings and metadata from the PDF, and pdfplumber extracts tables and converts them to Markdown.
2. **Chunking and embedding.** The text is split at section headings into 600-character chunks with 100 overlap. Tables stay whole, or are split by rows with the header repeated. Each chunk carries doc_id, version, page, section and chunk_type. Chunks are embedded with bge-small-en-v1.5 (384-dim, normalized) and stored in ChromaDB.
3. **Retrieval.** The question is embedded and ChromaDB returns the top 4 chunks by cosine similarity, with an optional version filter (v1 or v2).
4. **Refusal gate.** If the best similarity is below 0.68, or the keyword overlap between the question and the retrieved text is below 0.35, the system answers "Not found in the provided document."
5. **Generation.** The retrieved chunks are placed in an untrusted `<CONTEXT_CHUNKS>` block, separate from the system instructions. The prompt requires answers grounded only in that context, citations in the form [doc_id, version, page N, section], and a confidence level. Qwen2.5:3b runs locally through Ollama.
6. **Verification and logging.** A post-check confirms that every cited page was among the retrieved chunks and that key terms appear in them. The query, chunk IDs, answer, citations, confidence and verification status are saved to the SQLite audit log. The UI shows the answer, citations and a confidence badge.

Figure : Ingestion and query workflow



5. Evaluation Steps and Results

Describe the test question set, ground truth and metrics. Report actual results, including failures.

#	Test question / scenario	Expected answer / behaviour	Result (Pass/Fail, notes)
1	ASIL level and execution period of DoorLockSWC? (Factual)	ASIL-B, 20 ms	Pass. Page 5 at rank 1, cited.
2	Cycle time, bus and destination of Sig_CrashNotification? (Table lookup)	10 ms, CAN-FD, CanComSWC to DoorLockSWC	Pass. Page 8 at rank 1, High confidence, 4.8 s.
3	At speed above 15 km/h, which SWC receives it and what solenoid action follows? (Multi-hop)	DoorLockSWC via r_VehicleSpeed, 120 ms lock pulse on all 4 doors	Pass. Page 11 at rank 1, flow combined correctly.
4	Compare safety level and cycle time of CanComSWC and ImmobilizerSWC. (Comparison)	CanComSWC: ASIL-B, 10 ms. ImmobilizerSWC: QM, 100 ms	Pass. Page 5 at rank 1, both values correct.
5	Battery cooling pump PWM duty cycle? (Unanswerable)	"Not found in the provided document."	Pass. Both gates triggered, refused in 104 ms.

Metric	Score / observation
e.g. Answer relevance / faithfulness	Groundedness 100% (16/16 answerable questions were supported by the retrieved text). Refusal accuracy 100% (4/4 unanswerable questions refused, 0 false refusals).
e.g. Retrieval hit rate / citation accuracy	Hit@4 100% (16/16), Hit@1 81.2% (13/16), MRR 0.906. Citation validity 100% (every cited page was among the retrieved chunks).
e.g. Latency	Mean 4.04 s, minimum 0.10 s (refusals), maximum 5.76 s (multi-hop), measured on local CPU hardware.

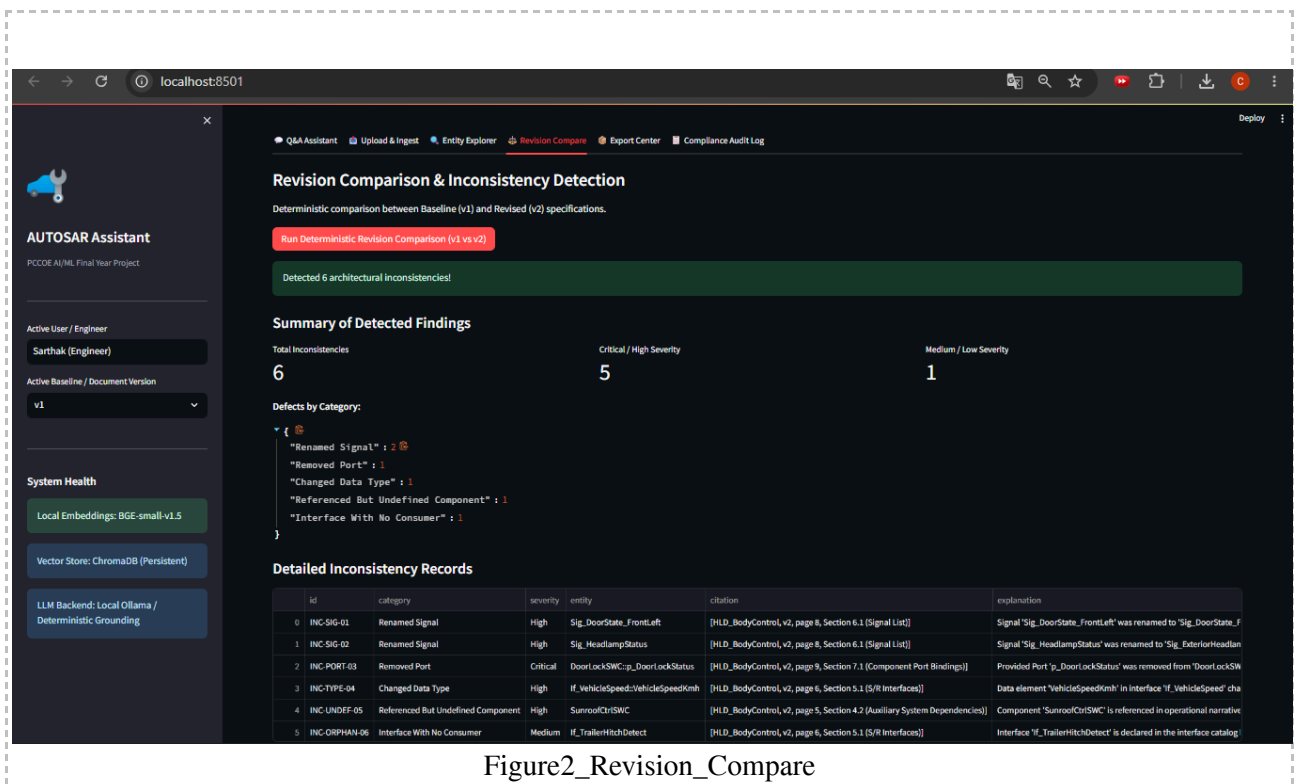
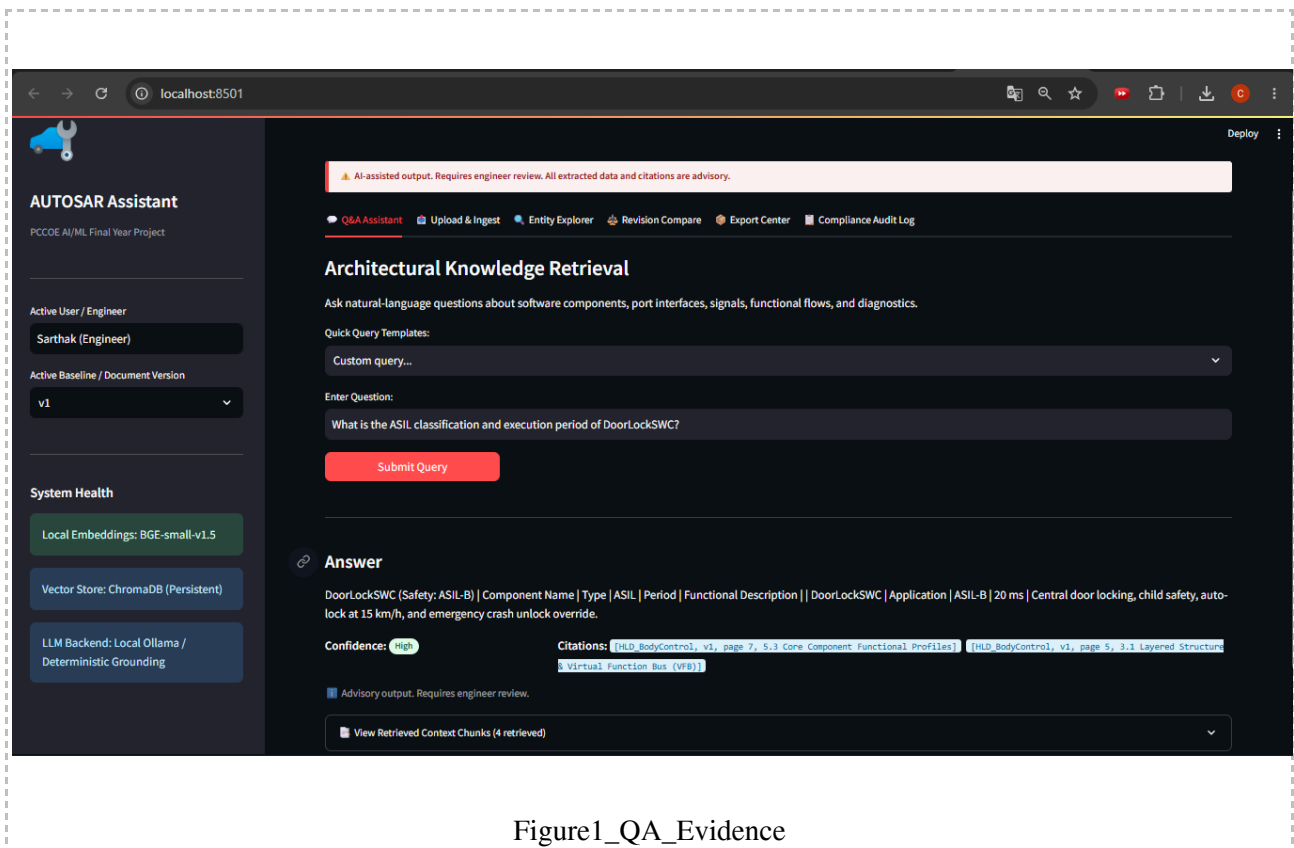
6. Responsible AI Measures

Area	Measure adopted in this project
Grounding and citations	Answers must cite sources as [doc_id, version, page, section]. A post-check downgrades or rejects answers whose citations were not in the retrieved context. A two-stage refusal gate returns "Not found in the provided document" when evidence is weak.
Privacy	The system runs fully offline: a local Qwen2.5:3b model (Ollama) and local embeddings. Documents, queries and answers stay on the host machine, and no external API is called. Only synthetic documents were used.
Security	Retrieved document text is placed in a separate <code><CONTEXT_CHUNKS></code> block and treated as untrusted data, not instructions, to reduce prompt-injection risk.
Human oversight / review	Every screen shows "AI-assisted output. Requires engineer review." The tool cannot approve or sign off changes. Queries, retrieved chunk IDs, answers and verification status are saved in a SQLite audit log.

7. Innovation Highlights

- Two-stage refusal gate.** A similarity threshold (0.68) plus a keyword-overlap check (0.35) makes the system answer "Not found in the provided document" instead of guessing. Refusals take about 0.1 s.
- Rule-based revision comparator.** Five deterministic rules detect renamed signals, removed ports, data type changes, undefined components and orphaned interfaces, and found all 6 seeded defects. The LLM only explains the findings.
- Section-aware chunking that keeps tables intact.** Tables are extracted as Markdown, and when a table is split the header row is repeated, so column context is not lost.
- Fully local pipeline.** The model, embeddings, vector store and UI run on a single laptop with no cloud dependency or API cost.

8. Screenshots, Logs and Evidence



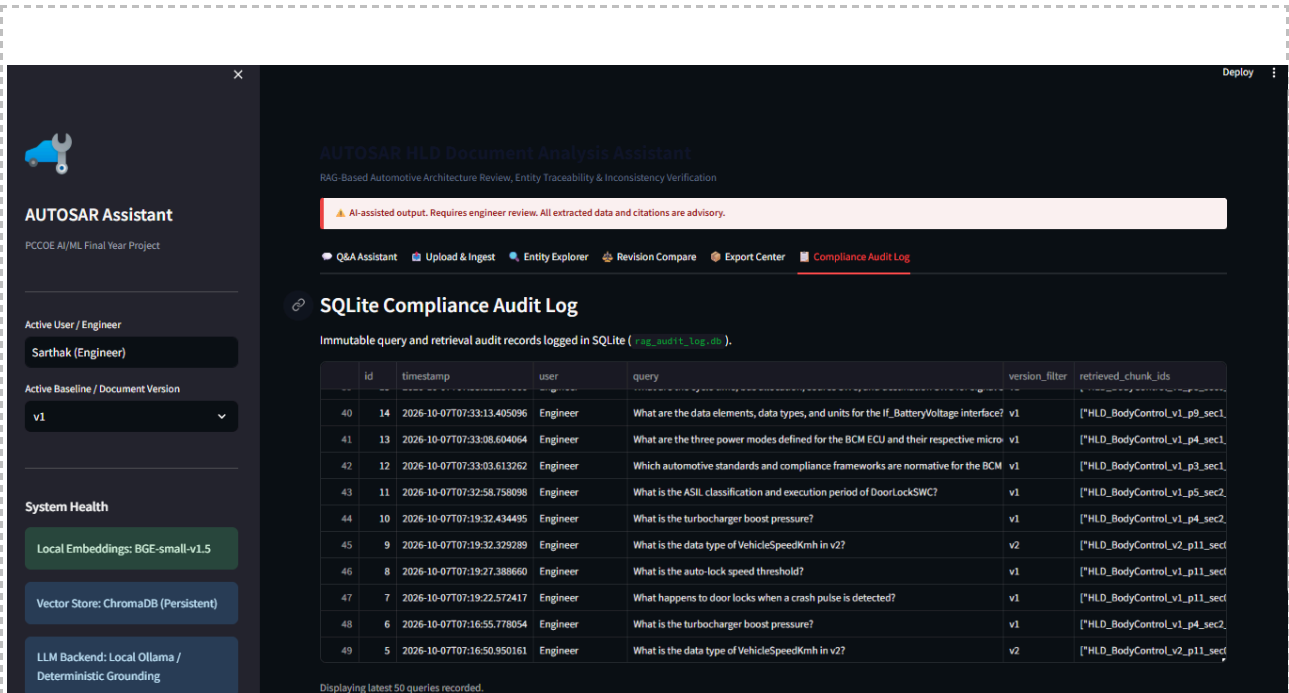


Figure3_Audit_Log_Evidence

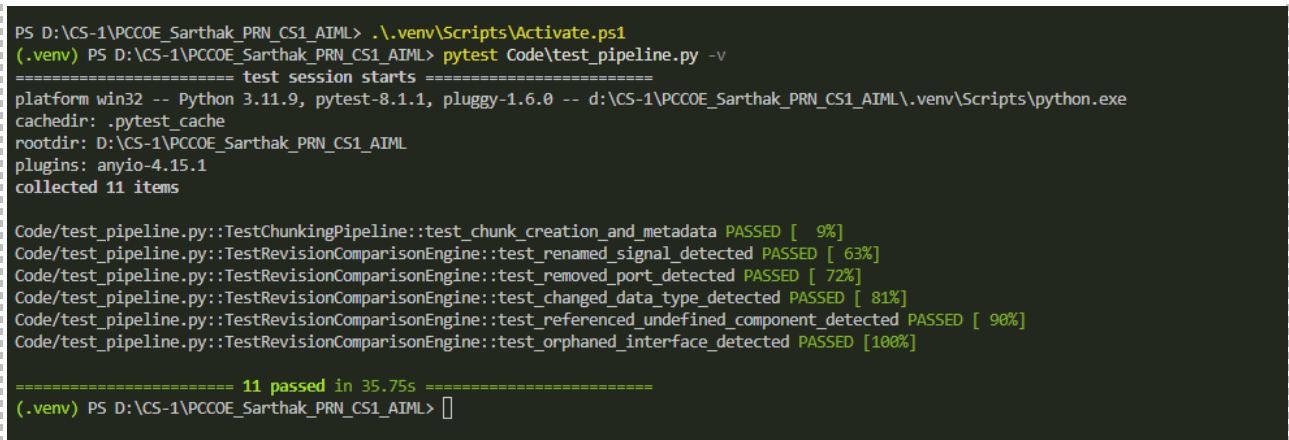


Figure4_Automated_Test_Evidence

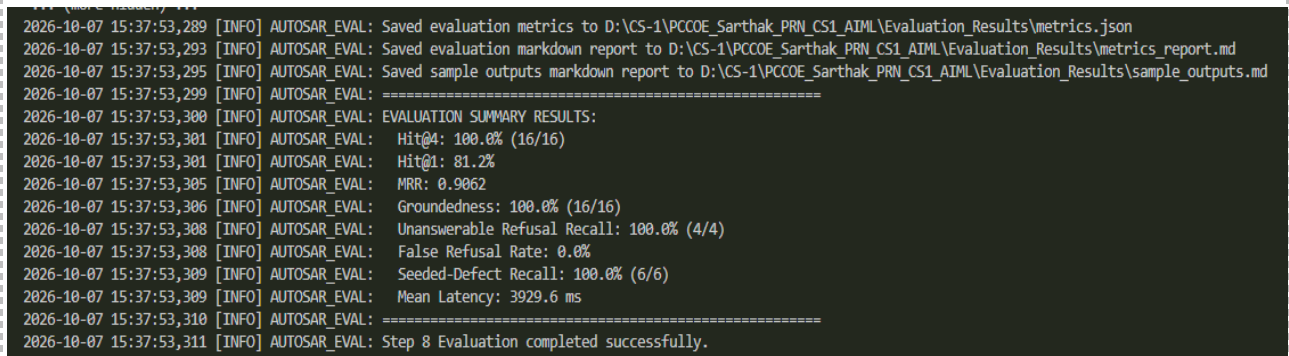


Figure5_Benchmark_Evaluation_Evidence

9. Reflection and Conclusions

Reflection

Combining deterministic rules with an LLM worked well. The LLM writes readable, cited answers, while rules reliably detect changes such as renamed signals and removed ports. Cosine similarity alone could not separate relevant from irrelevant questions, so I added a keyword-overlap check. The main limitations are the small synthetic test set and the lack of OCR, diagram and ARXML support.

Conclusion

The assistant is a working, fully local RAG prototype. On the 20-question test set it retrieved the right page in the top 4 for all 16 answerable questions (MRR 0.906), refused all 4 unanswerable ones, and detected all 6 seeded defects. Future work includes a larger test set, an API layer, access control and OCR.

10. AI-Tool Usage Declaration

AI tool / model	Purpose	Artifact affected	Verified By Student (Y/N)
Google Antigravity	Generate and Debugged the project code: ingestion, chunking, RAG pipeline, entity extraction, comparator, Streamlit UI, evaluation script	Code/	Y
Claude (Anthropic)	Planned the project scope and drafted the build Prompt	Build prompt only, no submitted code	Y
Qwen2.5:3b (via Ollama)	Runtime model that generates answers inside the system, not a development aid	System runtime	Y

11. Student Declaration

I confirm that this report represents my own understanding and contribution, that the work follows the approved case-study scope, and that I can explain every significant part of the implementation.

Signature:		Date:	
Name:	Sarthak Shivaji Bagul	PRN:	124B2B021